

$ident, x$		
exp, e	$::=$	
		x
		$()$
		(exp, exp')
		function $x \rightarrow exp$ bind x in exp
		$exp\ exp'$
		let $pat = exp$ in exp' bind $bo(pat)$ in exp'
		$\{exp/x\}exp'$ M
		$\sigma\ exp$ M
		(exp) M
val, v	$::=$	
		x
		$()$
		(val, val')
		function $x \rightarrow exp$
pat, p	$::=$	
		x $bo = x$
		$()$ $bo = \{\}$
		(pat, pat') $bo = bo(pat) \cup bo(pat')$
σ	$::=$	multiple term substitution
		$[x \mapsto e]$
		$[]$
		σ_1, σ_2
$formula$	$::=$	
		$judgement$
		not ($formula$)
$terminals$	$::=$	
		$\rightarrow$
		$\longrightarrow$
		$\mapsto$
Jop	$::=$	
		$e \longrightarrow e'$
		match (p, v) = σ
$judgement$	$::=$	
		Jop
$user_syntax$	$::=$	
		$ident$
		exp
		val
		pat
		σ

	<i>formula</i>
	<i>terminals</i>

$e \longrightarrow e'$

$$\begin{array}{c}
\frac{}{(\mathbf{function} \ x \rightarrow e) \ v \longrightarrow \{v/x\}e} \quad \text{JO_RED_FUN} \\
\\
\frac{\mathbf{match} \ (p, v) = \sigma}{\mathbf{let} \ p = v \ \mathbf{in} \ e \longrightarrow \sigma \ e} \quad \text{JO_RED_LET} \\
\\
\frac{e_1 \longrightarrow e'_1}{e_1 \ e_2 \longrightarrow e'_1 \ e_2} \quad \text{JO_RED_CONTEXT1} \\
\\
\frac{e_2 \longrightarrow e'_2}{v_1 \ e_2 \longrightarrow v_1 \ e'_2} \quad \text{JO_RED_CONTEXT2} \\
\\
\frac{e_1 \longrightarrow e'_1}{(e_1, e_2) \longrightarrow (e'_1, e_2)} \quad \text{JO_RED_CONTEXT3} \\
\\
\frac{e_2 \longrightarrow e'_2}{(v_1, e_2) \longrightarrow (v_1, e'_2)} \quad \text{JO_RED_CONTEXT4} \\
\\
\frac{e_1 \longrightarrow e'_1}{\mathbf{let} \ p = e_1 \ \mathbf{in} \ e_2 \longrightarrow \mathbf{let} \ p = e'_1 \ \mathbf{in} \ e_2} \quad \text{JO_RED_CONTEXT5}
\end{array}$$

$\mathbf{match} \ (p, v) = \sigma$

$$\begin{array}{c}
\frac{}{\mathbf{match} \ (x, v) = [x \mapsto v]} \quad \text{JO_M_IDENT} \\
\\
\frac{}{\mathbf{match} \ ((), ()) = []} \quad \text{JO_M_UNIT} \\
\\
\frac{\mathbf{match} \ (p_1, v_1) = \sigma_1 \quad \mathbf{match} \ (p_2, v_2) = \sigma_2}{\mathbf{match} \ ((p_1, p_2), (v_1, v_2)) = \sigma_1, \sigma_2} \quad \text{JO_M_PAIR}
\end{array}$$

Definition rules: 10 good 0 bad
 Definition rule clauses: 18 good 0 bad